BE Minor Project Final Defence

TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

PULCHOWK CAMPUS

Global MPPT for PV Systems under Partial Shading Conditions using PSO with Tuned PI Controller

Submitted by

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1. Introduction

- **Nepal:** hydropower-dominated grid, weak in dry winters; ~4.7 kWh/m²/day insolation and ~300 sunny days a year
- **PV output is nonlinear:** one Maximum Power Point (MPP) that shifts with irradiance
- **Need:** a DC-DC converter + MPPT keeps the array at its MPP and extracts the most energy
- **Real rooftops get shaded:** the P-V curve gets several peaks and simple trackers fail

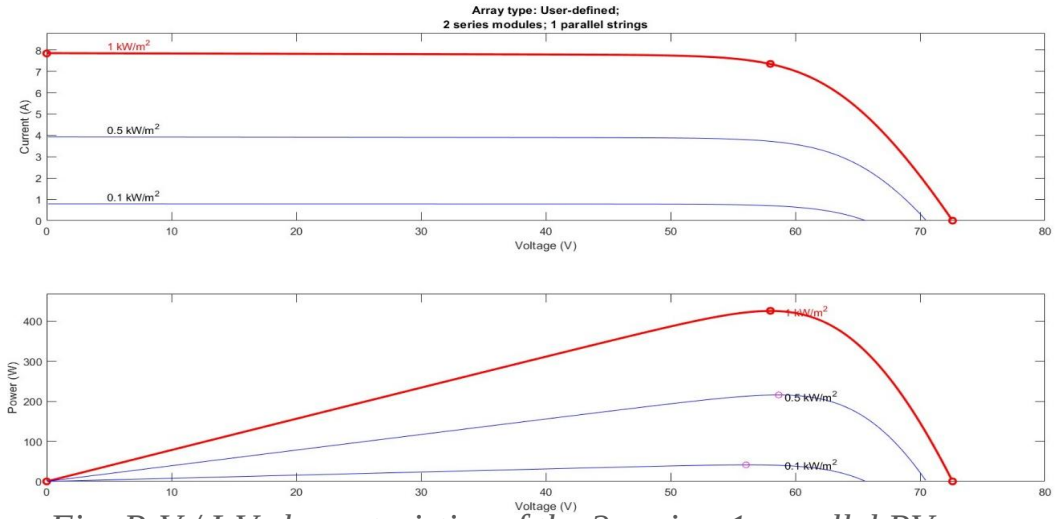


Fig: P-V / I-V characteristics of the 2-series, 1-parallel PV array

2. Problem Statement

- **P&O step-size trade-off:** large step = fast but oscillates around the MPP; small step = smooth but slow. A PI loop on the voltage reference can give both.
- **Partial shading:** bypass diodes create several local peaks plus one Global MPP (GMPP). Hill-climbing cannot tell them apart and stalls at a fraction of the available power.
- **What is needed:** a global search (PSO) to find the GMPP, plus a tuned PI controller for low-ripple regulation.

3. Objectives

Main focus: global MPPT of PV under partial shading using PSO with a tuned PI controller.

- Model a PV array with a DC-DC converter in MATLAB/Simulink (25 °C, varying irradiance)
- Implement P&O with PI control; evaluate under uniform and step-varying irradiance
- Simulate partial shading and show P&O failing to find the GMPP
- Design a PSO-based global MPPT and test its convergence to the GMPP
- Compare P&O and PSO schemes

4. Scope and Assumptions

Considered

- Simulation only (MATLAB/Simulink)
- 2S1P PV array + buck converter, DC side
- Step-varying irradiance; 4 modules in series for shading
- P&O + PI vs PSO + PI

Neglected / Assumed

- No DC-AC inverter, grid sync or power quality
- No hardware prototype
- Constant 1 Ω resistive load
- Cell temperature fixed at 25 °C
- PI gains and PSO parameters tuned by trial, not exhaustive search

5. System Model

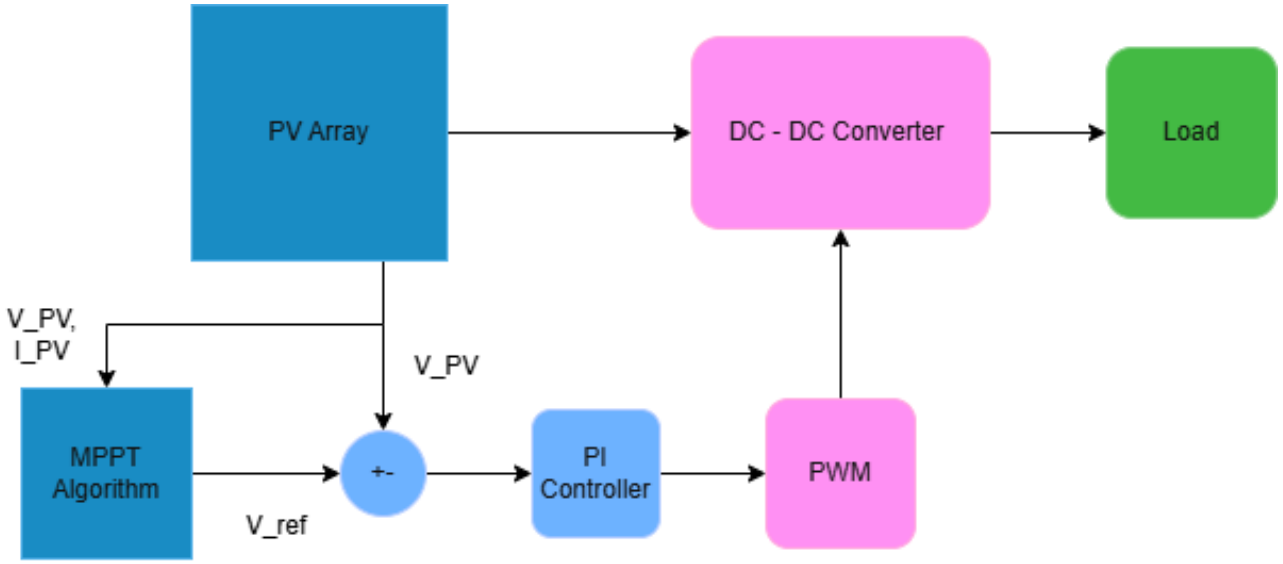


Fig: Closed-loop PV - MPPT - Buck converter system in Simulink

- **PV array (1000 W/m², 25 °C):** Voc ≈ 72 V, Isc ≈ 7.9 A, Pmax ≈ 425 W at ≈ 58 V
- **Buck converter:** Cin = Cout = 1000 µF, L = 100 mH, R = 1 Ω; D ≈ 0.35 at MPP
- **Control:** MPPT gives Vref; PI acts on (Vref - V_PV); PWM sets the duty cycle

6. P&O with PI Controller

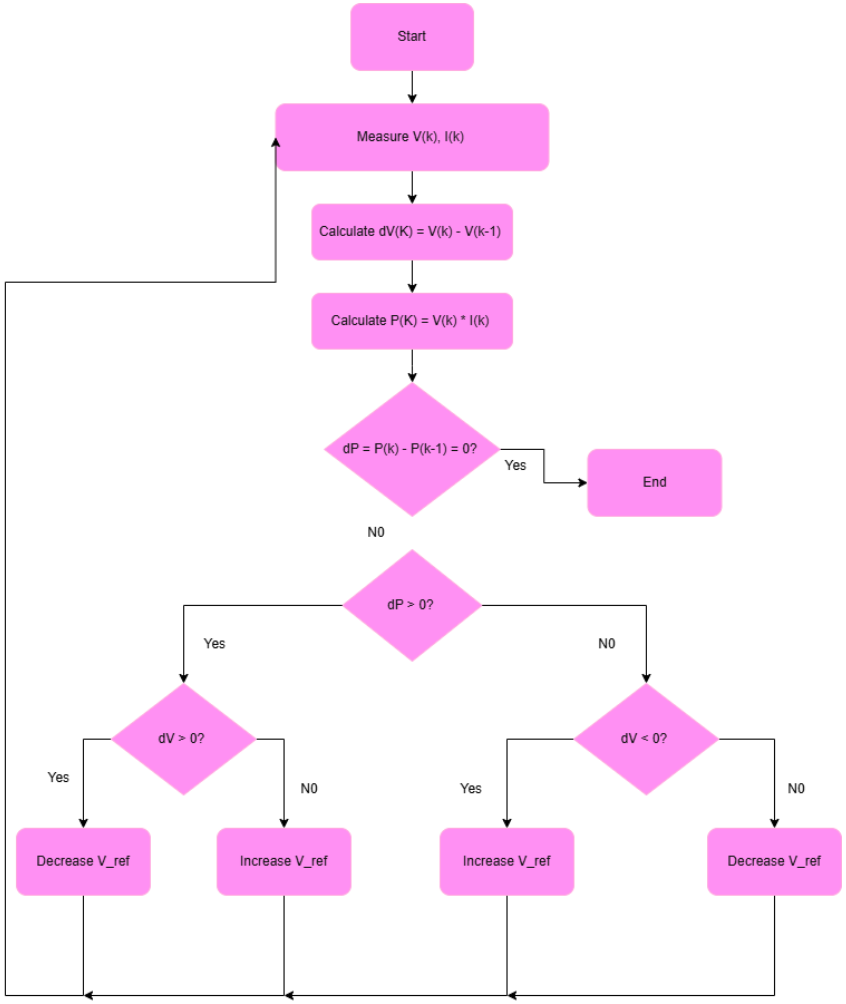


Fig: P&O flowchart

- Sample V and I each cycle; compute P
- If P rises, keep the direction; otherwise reverse
- Vref step = 10 V, limited to 10-70 V
- PI regulates V_PV towards Vref and sets the duty cycle
- Kp, Ki chosen by trial and error (several pairs compared)
- Also run without PI for comparison

7. PSO-based Global MPPT

- Each particle = a candidate Vref; N = 4, spaced over 5-70 V
- Particles evaluated one per sampling instant; pbest and gbest updated

$$v(k+1) = w \cdot v(k) + c_1 r_1 (pbest - x(k)) + c_2 r_2 (gbest - x(k))$$
$$x(k+1) = x(k) + v(k+1)$$

- w = 0.5, c_1 = 1.5, c_2 = 2.0; r_1, r_2 drawn from U[0, 1]
- Velocity limited to 5 V per iteration; position bounded to 5-70 V
- Swarm iterates continuously; no restart on irradiance change

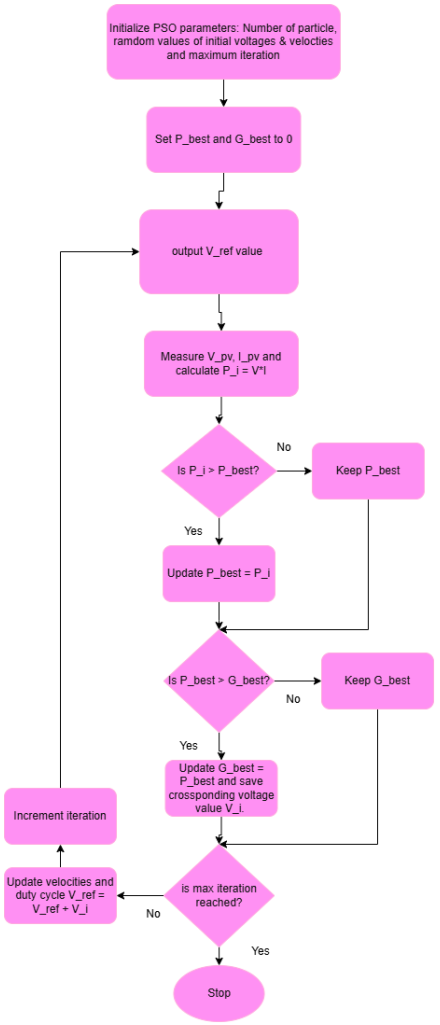
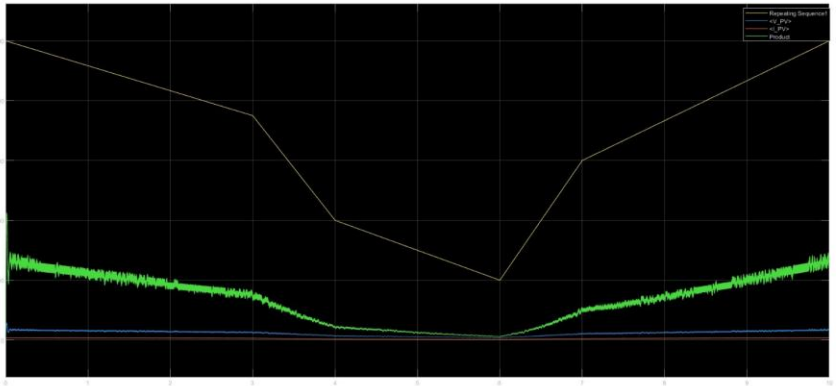
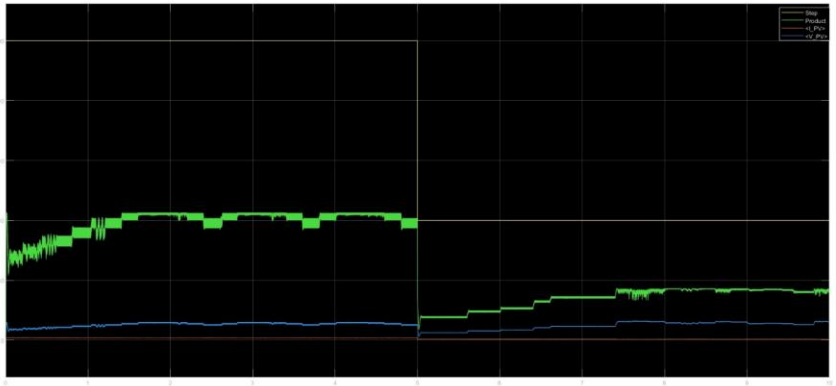


Fig: PSO flowchart

8. Results: Why MPPT is Needed



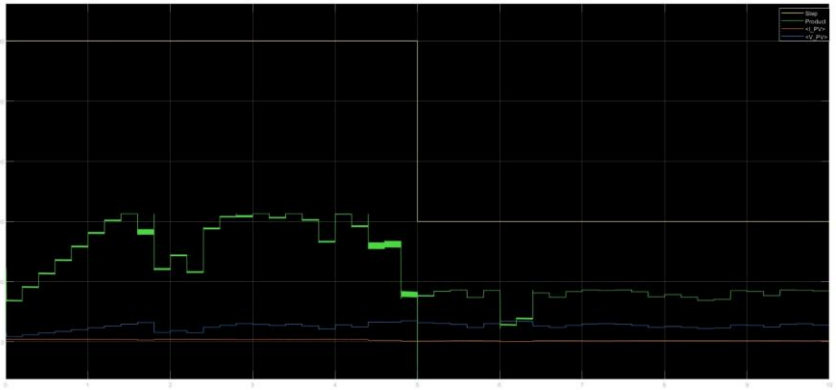
Fixed duty cycle



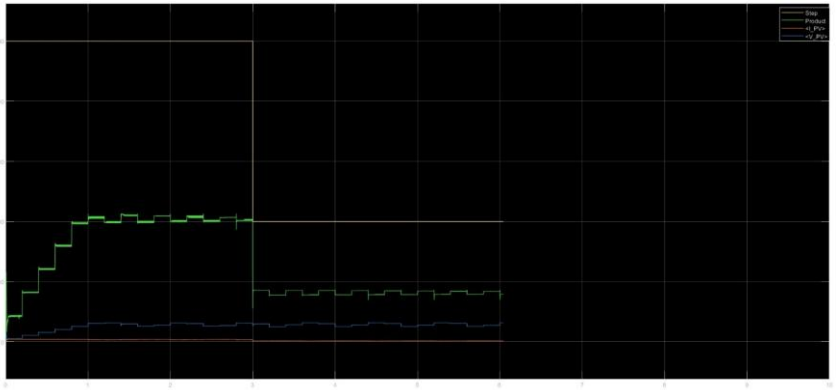
P&O without PI

- Fixed duty cycle: only ≈ 250-270 W of ≈ 425 W at 1000 W/m²; near 0 W at 200 W/m²
- P&O without PI (step 0.01): reaches ≈ 420 W but slowly (≈ 1.5 s; ≈ 2.5 s after the drop to 400 W/m²) and oscillates at the MPP

9. Results: PI Controller Tuning



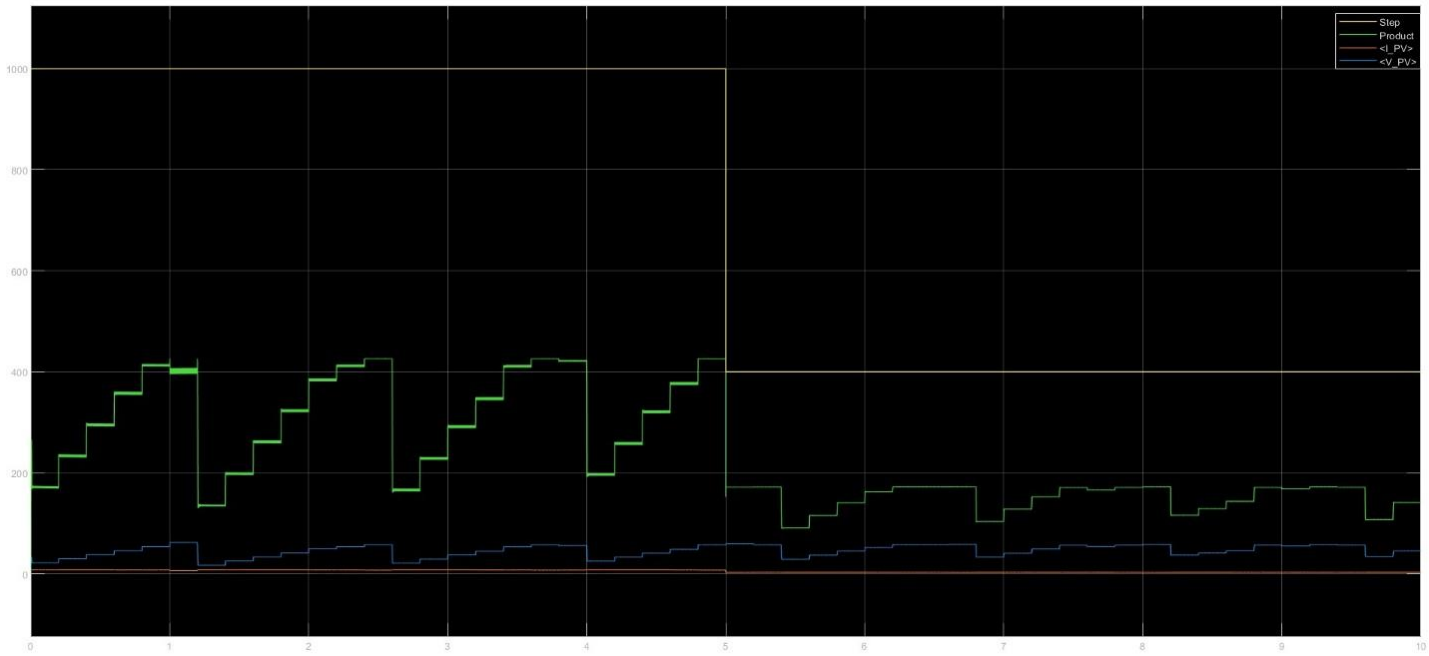
Kp = 1, Ki = 1



Kp = 0.2, Ki = 20

- Kp = 1, Ki = 1: irregular response, power repeatedly falls far below the MPP and never settles
- Kp = 0.1, Ki = 5 and Kp = 0.2, Ki = 20: ≈ 420 W in ≈ 1 s, quick recovery after the step to 400 W/m², small regular ripple

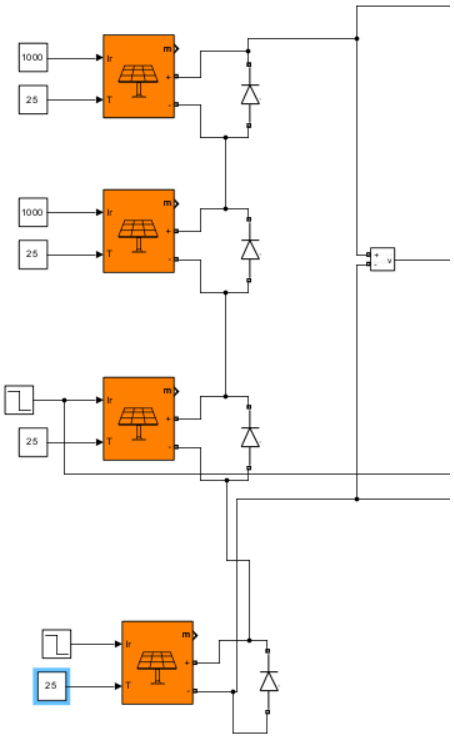
10. Results: PSO, Uniform Irradiance



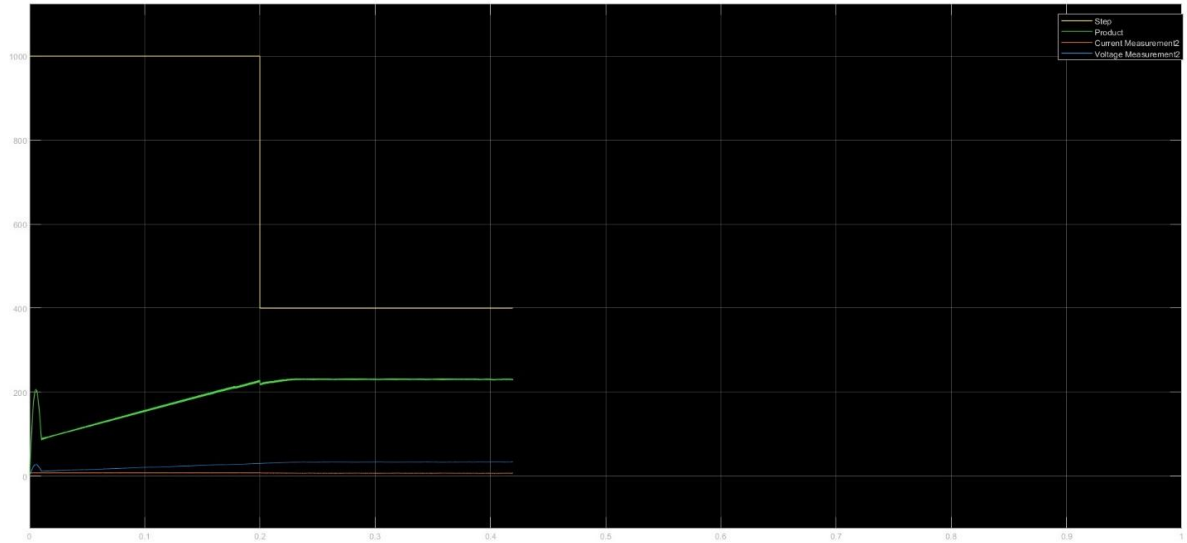
PSO under step-varying irradiance

- Reaches ≈ 420 W within ≈ 1 s
- Swarm keeps exploring: saw-tooth dips to ≈ 130-200 W waste energy, so P&O with tuned PI is better under uniform irradiance

11. Results: P&O under Partial Shading



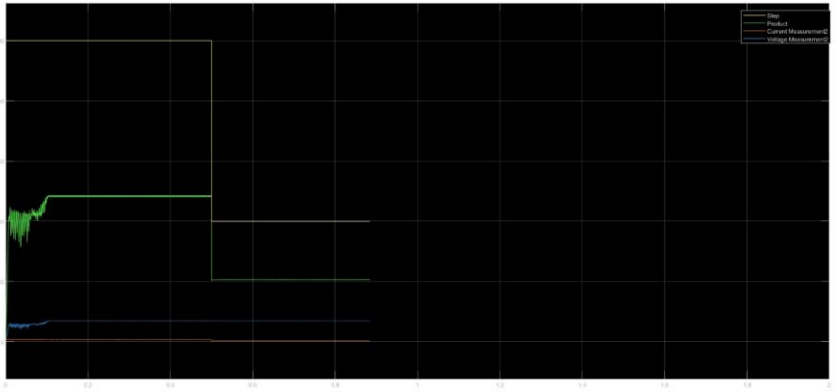
Shading setup



P&O with PI

- Without PI: ≈ 440 W, then drops to 0 W when shading starts
- With PI: stuck at ≈ 230 W (≈ 35 V) on the first peak
- ≈ 485 W was available: P&O only follows the local gradient

12. Results: PSO under Partial Shading



Kp = 0.5, Ki = 20 (best)



Ki = 1045 (unstable)

- Converges in ≈ 0.1 s to ≈ 485 W at ≈ 70 V with negligible ripple (P&O: ≈ 230 W)
- Best gains by trial: Kp = 0.5, Ki = 20; very high Ki makes the loop oscillate
- After shading it holds ≈ 205 W: a restart trigger is needed to find the new GMPP

13. Results: P&O vs PSO

Criterion	P&O + PI	PSO + PI
Uniform irradiance	≈ 420 W in ≈ 1 s; small, regular ripple	≈ 420 W in ≈ 1 s; saw-tooth dips to ≈ 130-200 W
Partial shading	Stuck on first peak: ≈ 230 W	Finds GMPP: ≈ 485 W in ≈ 0.1 s
Ripple at steady state	Small (tuned PI)	Negligible after convergence
Main limitation	Local search; cannot find the GMPP	No restart on irradiance change

Best of both: PSO locates the global region, then P&O fine-tunes the peak.

14. Conclusion

- A fixed duty cycle wastes power, so MPPT is necessary
- Uniform irradiance: P&O with tuned PI (Kp = 0.2, Ki = 20) is smooth and fast
- Partial shading: P&O stalls at ≈ 230 W; PSO reaches ≈ 485 W in ≈ 0.1 s
- Learned: PI tuning matters (very high Ki is unstable) and PSO needs a restart trigger
- Next: hybrid PSO-P&O, restart on irradiance change, hardware validation

References

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